The Newton cusp: classical histories, conditional floors and records

Maintained conclusions. A bounded continuous conservative force merges two full point states in finite time. Two different conditional preparation theorems then force a positive covariance action: one assumes reversible evolution on instantaneous phase laws, canonical covariance and weak closure; the other assumes actual Gaussian cusp transport, W_2 closure, copying and continuity at one positive time. Neither theorem identifies that floor with physical $\hbar/2$. Their classical density and entropy witnesses carry supplied scales.

There is also an exact zero-floor classical completion on complete trajectories. It preserves past and future branch information beyond the instantaneous phase point. Arbitrarily fine nondisturbing joint records force a zero covariance infimum. These countermodels delimit the preparation theorems; they do not contradict their explicitly different state and instrument assumptions.

The reversible theorem constrains point transport **only if points are admitted**. A positive-floor Gaussian class with identity evolution satisfies that clause vacuously and is not physical cusp transport. The continuity theorem below instead constrains actual nonpoint transport. Quantum existence at every supplied $\hbar > 0$, quantum instruments, and universality of a measured action constant remain separate questions.

The solution-space construction has prior context in Sell 1973 (publisher abstract). Vanishing classical uncertainty and branch probabilities belong to spontaneous stochasticity; see Eyink–Drivas 2015 and Vanpoucke–Wenmackers 2021 (author abstracts). The proofs below are written arguments using their stated analytic inputs; internal model checks are not independent peer review. No priority is claimed. The classical construction distinguishes actual mergers from an unbounded force slope or an endpoint caustic.

1. Existence without uniqueness, with no unbounded force

Fix $m, \gamma, R > 0$, $0 < \alpha < 1$, and a smooth even cutoff χ with $0 \le \chi \le 1$, $\chi = 1$ on $[-1, 1]$ and $\chi = 0$ outside $[-2, 2]$. Set

$$V(q) = -\frac{\gamma}{1+\alpha}|q|^{1+\alpha}\chi(q/R), \qquad \dot{q} = p/m, \quad \dot{p} = -V'(q). \tag{1}$$

The potential is bounded and C^1, and the force is continuous and bounded. Specifically,

$$\|V\|_\infty \le \frac{\gamma(2R)^{1+\alpha}}{1+\alpha}, \qquad \|V'\|_\infty \le \gamma(2R)^\alpha\left(1 + \frac{2\|\chi'\|_\infty}{1+\alpha}\right) =: F_*. \tag{2}$$

Inside $|q| < R$, the acceleration is $\ddot{q} = (\gamma/m)\,\mathrm{sgn}(q)|q|^\alpha$. Its derivative diverges at zero, so the Lipschitz hypothesis in the lesson fails. Nevertheless solutions exist globally. Here is the relevant existence argument without assuming uniqueness. Euler polygons on a fixed finite time interval have uniformly bounded momentum and position, and are equi-Lipschitz. On their common compact range the continuous vector field is uniformly continuous, so their integral-equation residual tends uniformly to zero with the mesh. A uniformly convergent subsequence therefore solves (1). The bounds

$$|p(t)| \le |p(0)| + F_*|t|, \qquad |q(t)| \le |q(0)| + \frac{|p(0)|}{m}|t| + \frac{F_* t^2}{2m} \tag{3}$$

prevent finite-time escape and allow successive continuation in either time direction. This is an existence construction by compactness, rather than the Lipschitz lesson's Cauchy construction of a unique limit.

Put

$$\beta = \frac{2}{1-\alpha} > 2, \qquad B = \left[\frac{\gamma(1-\alpha)^2}{2m(1+\alpha)}\right]^{1/(1-\alpha)}. \tag{4}$$

The curves $q(t) = \pm B(t-t_0)^\beta$, preceded by rest at zero, solve (1) while their positions remain inside $(-R, R)$. They join with two continuous derivatives at t_0. Rest and an arbitrary delay before either departure are therefore all permitted by the same initial datum $(q, p) = (0, 0)$. Direct substitution uses $\beta - 2 = \alpha\beta$ and $B^{1-\alpha} = \gamma/[m\beta(\beta-1)]$.

More decisively, choose $s > 0$ with $Bs^\beta < R$. The two different incoming states

$$z_\pm = (\pm Bs^\beta,\ \mp m\beta B s^{\beta-1}) \tag{5}$$

have the unique trajectories $q_\pm(t) = \pm B(s-t)^\beta$, $p_\pm = m\dot{q}_\pm$ for $0 \le t < s$: the vector field is locally Lipschitz wherever $q \neq 0$. Both reach $(0, 0)$ at time s. These are mergers of the full canonical state, rather than caustics of its position projection.

Point-state obstruction. No injective evolution at time s on the original (q, p) states can follow both trajectories through their arrival. A later choice of departure cannot undo the earlier merger. An extra arrival or branch label can retain the information, but is an enlargement of this state space.

For the path limits below write $k = 1 + \alpha$, $\lambda = \gamma/(1+\alpha)$, and $\Gamma_\pm(t) = (\pm Bt^\beta, \pm m\beta B t^{\beta-1})$ for $BT^\beta < R$. Fix calibration units $L_0, P_0 > 0$, $S_0 = L_0 P_0$, and $z = (q/L_0, p/P_0)$. These units do not impose a physical bound.

2. The trajectory-state completion

Let Ω be the set of all globally defined solutions of the bounded cusp equation, viewed as functions $\gamma : \mathbb{R} \to \mathbb{R}^2$ with the topology of uniform convergence on compact intervals. Write $e_t(\gamma) = \gamma(t)$ and

$$(T_s\gamma)(t) = \gamma(t+s), \qquad (\Theta\gamma)(t) = \big(q(-t), -p(-t)\big). \tag{H4}$$

Theorem 1. Ω is a nonempty Polish space. The maps T_s form a jointly continuous group of homeomorphisms and Θ is a continuous involution satisfying $\Theta T_s \Theta = T_{-s}$. Every e_0-fiber is nonempty and compact. The state class $\mathcal{P}(\Omega)$ of all Borel probabilities has weakly continuous reversible evolution $\mu \mapsto (T_s)_*\mu$, contains point states, and supports ordinary classical conditioning on any Borel trajectory record of positive probability. Its instantaneous phase laws comprise all Borel probabilities on $\mathbb{R}^2$; in particular their covariance-action infimum is zero.

Proof. The integral form of the bounded cusp equation is preserved by local-uniform limits: on each compact time interval the convergent paths have a common compact range, and the continuous vector field is uniformly continuous there. Thus Ω is closed in the Polish compact-open space $C(\mathbb{R}, \mathbb{R}^2)$. Time translations and time reversal preserve the autonomous equation and have the asserted algebra and continuity.

Given a compact set of initial points, the bounded-force estimates above give locally uniform bounds on q, p and their derivatives. Arzelà–Ascoli, a diagonal subsequence and closedness give compactness of $e_0^{-1}(K)$. Global existence makes every fiber nonempty. Pushforward by the continuous time shifts gives the group on $\mathcal{P}(\Omega)$; for each bounded continuous test function, dominated convergence proves weak continuity in time. Ordinary restriction and normalization give the stated conditioning rule.

For the marginal claim, the proper continuous surjection e_0 has a Borel section $s : \mathbb{R}^2 \to \Omega$. One justification is the Kuratowski–Ryll-Nardzewski measurable-selection theorem (statement, theorem statement read): the compact-valued fiber map is Borel-measurable, since the image under a proper map of a closed set is closed and an open set in a metric space is a countable

union of closed sets. Hence $s_*\nu$ is a history preparation with instantaneous marginal ν for any $\nu \in \mathcal{P}(\mathbb{R}^2)$. A Dirac history has a Dirac instantaneous marginal, with $\kappa = 0$. □

The theorem constructs reversible *history* evolution, not a reversible map on laws of (q, p) alone. Take two histories that arrive at the cusp from opposite sides. At their common arrival time their instantaneous marginals are both $\delta_{(0,0)}$, while the full states and their pasts remain distinct. The resting history and a departing history have the same instantaneous marginal at time zero but different future marginals, so no single map on phase laws can represent the shift for all history states. Moreover, the resting solution and the two solutions which leave after any prescribed delay $a \geq 0$ have the same entire (q, p) past up to time zero but different futures. Thus a deterministic completion representing all these possibilities must carry information beyond that past. In (H4) the chosen future is part of the state. The construction does not show that this information has a finite-dimensional, local Hamiltonian realization or is accessible to a physical observer.

Because the phase marginal class is all of $\mathcal{P}(\mathbb{R}^2)$, it is weakly closed, affine-canonically invariant, and closed under classical product preparations, canonical mixing and marginalization **as a class of single-time laws**. The Borel section is a state-preparation selection; it is not a proof of a reversible canonical mixer acting on the enlarged history states. Nor does the instantaneous marginal determine its later law. The preparation-floor theorem explicitly assumes that a reversible evolution acts on those marginals themselves; Theorem 1 does not satisfy that premise, so the two results are consistent.

3. A continuous classical family with nonuniform zero-width limit

Let X, Y be independent standard normal variables. For fixed $u \in \mathbb{R}$ and $\epsilon > 0$, prepare

$$q_0 = \epsilon L_0(u + X), \qquad p_0 = \epsilon P_0 Y. \tag{H5}$$

The energy-zero set is Lebesgue-null, so (H5) selects unique global trajectories almost surely. Equivalently, lift the law by the section s: different choices of s agree almost surely here. Let Φ denote the standard normal distribution function.

Theorem 2. Let $0 < T < (R/B)^{1/\beta}$. As $\epsilon \to 0$, the initial law of (H5) converges weakly to $\delta_{(0,0)}$ for every fixed u, and

$$\kappa_\epsilon = S_0\epsilon^2, \qquad \mathrm{Law}\big((q(t), p(t))_{0\leq t\leq T}\big) \Longrightarrow \Phi(u)\,\delta_{\Gamma_+} + \Phi(-u)\,\delta_{\Gamma_-} \quad \text{in } C([0, T], \mathbb{R}^2). \tag{H6}$$

Every mixture weight strictly between zero and one is obtained by a fixed u. For either pure branch one may instead take $u = u_\epsilon \to \pm\infty$ with $\epsilon|u_\epsilon| \to 0$. The distributions are classical ensembles of Newton trajectories, with no supplied $\hbar$.

Proof. The first limit and covariance are immediate. Fix a realization with $u + X \neq 0$ and finite Y. Its initial energy is, for small ϵ,

$$E_\epsilon = \frac{\epsilon^2 P_0^2 Y^2}{2m} - \lambda\,|\epsilon L_0(u + X)|^k < 0. \tag{H7}$$

Indeed $k < 2$. Thus it cannot reach $q = 0$, so its sign remains that of $u + X$. Its turning distance $a_\epsilon = (-E_\epsilon/\lambda)^{1/k}$ satisfies $0 < a_\epsilon \leq |q_0|$, and tends to zero. Inside $|q| < R$, separation of the energy equation gives the time from the turning point to $r \geq a$:

$$I_a(r) = \sqrt{\frac{m}{2\lambda}} \int_a^r \frac{dv}{\sqrt{v^k - a^k}}. \tag{H8}$$

For fixed $0 < r < R$, $I_a(r) \to I_0(r)$ as $a \downarrow 0$, where

$$I_0(r) = \sqrt{\frac{m}{2\lambda}} \frac{2}{1-\alpha} r^{(1-\alpha)/2}, \qquad I_0^{-1}(t) = Bt^\beta. \tag{H9}$$

For the convergence, the part from a to $2a$ is $O(a^{(1-\alpha)/2})$; on $v \geq 2a$ the integrand is dominated by a constant times $v^{-k/2}$, which is integrable at zero. Moreover,

$$I_a(r) \leq \sqrt{\frac{m}{2\lambda}} \frac{2}{1-\alpha} (r-a)^{(1-\alpha)/2}, \tag{H10}$$

since $v^k - a^k \geq (v-a)^k$ for $k > 1$. Hence the time from the initial point to its turning point is $I_{a_\epsilon}(|q_0|) \to 0$. The outward time to any fixed r is $I_{a_\epsilon}(r)$ plus or minus this vanishing initial term. Bracketing fixed times by the hitting times of two nearby radii proves $q(t) \to \operatorname{sgn}(u+X)Bt^\beta$ for each $t > 0$. The energy equation gives the corresponding momentum limit, with outward sign. For T as stated the limiting path remains inside $|q| < R$; the same hitting-time argument keeps the approximating paths there for all sufficiently small ϵ.

The bounded-force estimates give equicontinuity of both coordinates on $[0, T]$, so the pointwise convergence is uniform there. It holds for almost every (X, Y), and bounded convergence of continuous path functionals gives (H6). If $u_\epsilon \to \pm\infty$ with $\epsilon|u_\epsilon| \to 0$, the same argument has eventual fixed sign for each (X, Y) and gives the corresponding pure branch. □

Fixed-width continuity. Let $\mu_{z,\epsilon} = s_* \mathcal{N}(z, \epsilon^2 I)$ on Ω, with z dimensionless. For all centers z, w and every t,

$$\left\| (T_t)_* \mu_{z,\epsilon} - (T_t)_* \mu_{w,\epsilon} \right\|_{\mathrm{TV}} \leq \frac{|z-w|}{\sqrt{2\pi}\,\epsilon}. \tag{H11}$$

Here total variation is the supremum over measurable events. Pushforward cannot increase it. For the Gaussian bound, interpolate its center along the line from z to w and integrate the absolute directional derivative: half its integral is $1/(\sqrt{2\pi}\epsilon)$ per unit center distance. Equation (H11) applies to any subsequent fixed measurable record, including a complete history record. Its constant diverges as $\epsilon \downarrow 0$.

The divergence cannot be removed uniformly. Fix $0 < t < T$ and a bounded continuous f with $f(Bt^\beta) = 1$ and $f(-Bt^\beta) = 0$; a clipped linear function suffices. For the two centers $z_\pm = (\pm\epsilon, 0)$, Theorem 2 gives

$$\lim_{\epsilon \downarrow 0} \left[\mathbb{E}_{z_+,\epsilon} f(q(t)) - \mathbb{E}_{z_-,\epsilon} f(q(t)) \right] = 2\Phi(1) - 1 > 0. \tag{H12}$$

The two initial centers are only 2ϵ apart. Thus no common modulus of continuity tending to zero can hold for these responses as ϵ varies. A uniform response modulus would exclude the zero-width sequence; it is an extra physical premise, not a consequence of continuity at each fixed width. More concretely, if the absolute response difference is bounded by $L|z-w|$ for some $L > 0$, uniformly over all allowed widths, and both translated preparations at each such width are admitted, (H12) implies an $\epsilon_0 > 0$ such that widths below $\min\{\epsilon_0, [2\Phi(1)-1]/(4L)\}$ are excluded for this family. This conditional bound does not identify $S_0\epsilon^2$ with $\hbar$ and does not establish the response premise for physical records.

Corollary. No evolution on phase-space probability laws which agrees with the actual Newton transport of all the Gaussian laws (H5) can be weakly continuous at $\delta_{(0,0)}$ at any fixed $t \in (0, T)$. Indeed all the inputs have that same weak limit while (H6) gives distinct output limits for different u. This obstruction does not require injectivity or a rule for the exact point at the origin. It does require the indicated Gaussian preparations and their actual cusp transport.

4. Reversible instantaneous preparations

Let $\mathcal{S}$ be a nonempty subset of probability measures on the canonical plane, containing at least one measure with finite covariance. Assume the following three properties.

1. **Canonical covariance of preparations.** Every affine symplectic pushforward of a member of $\mathcal{S}$ belongs to $\mathcal{S}$. In one canonical pair the linear matrices have determinant one.

This is a premise about admissible states, not a claim that the particular potential (1) is invariant under all these maps.

2. **Weak closure.** A probability-valued weak limit of a sequence in $\mathcal{S}$ belongs to $\mathcal{S}$.
3. **Reversible point-trajectory compatibility.** The force (1) admits bijective preparation evolutions $\mathcal{E}_t$ on $\mathcal{S}$, with weakly continuous state curves. Whenever a point preparation is admitted, its evolution follows its unique Newton trajectory before arrival at the cusp. No affineness assumption on $\mathcal{E}_t$ is needed for the conclusion below.

For a finite-covariance preparation write

$$\kappa(\mu) = \sqrt{\det \mathrm{Cov}_\mu(q,p)}. \tag{6}$$

This has action units. The theorem concerns covariance, including its cross term, rather than the coordinate product alone.

Theorem R. Under properties 1–3,

$$0 < \kappa_* := \inf_{\substack{\mu\in\mathcal{S} \\ \mu \text{ has finite covariance}}} \kappa(\mu) < \infty. \tag{7}$$

First step: point preparations are excluded. If one Dirac measure belongs to $\mathcal{S}$, affine translations supply every Dirac measure, in particular δ_{z_+} and δ_{z_-}. Property 3 and weak time continuity then give

$$\mathcal{E}_s\delta_{z_+} = \delta_{(0,0)} = \mathcal{E}_s\delta_{z_-}, \tag{8}$$

contradicting injectivity. Thus $\mathcal{S}$ contains no Dirac measure.

Second step: arbitrarily small covariance action would supply a point. For clarity fix arbitrary calibration units $L_0, P_0 > 0$, with $S_0 = L_0P_0$, and use dimensionless coordinates $(Q,P) = (q/L_0, p/P_0)$. If (7) failed, choose preparations with dimensionless covariance matrices $C_n \geq 0$ and $\sqrt{\det C_n} = \kappa(\mu_n)/S_0 \to 0$. Subtract each mean. Choose $\epsilon_n > 0$ so that

$$\sqrt{\det D_n} \longrightarrow 0, \qquad D_n = C_n + \epsilon_n I > 0, \qquad A_n = (\det D_n)^{1/4} D_n^{-1/2}. \tag{9}$$

Such a choice follows from continuity of determinant at each C_n; it also handles rank-deficient covariances. Then $\det A_n = 1$ and

$$A_n C_n A_n^{\mathsf{T}} \leq \sqrt{\det D_n}\, I. \tag{10}$$

These dimensionless determinant-one maps represent physical canonical maps in the calibrated coordinates. By property 1 the centered, transformed preparations $\widetilde{\mu}_n$ remain admissible. For every fixed $r > 0$, Chebyshev's inequality gives

$$\widetilde{\mu}_n\{Q^2 + P^2 > r^2\} \leq \frac{2\sqrt{\det D_n}}{r^2} \longrightarrow 0. \tag{11}$$

Hence $\widetilde{\mu}_n$ converges weakly to the point measure at zero. Property 2 admits that point, contrary to the first step. The assumed finite-covariance member also makes the infimum finite. □

No calibrated numerical value is hidden in L_0, P_0: their positive values are arbitrary and (7) is unchanged by changing them. The state class supplies the value of κ_*. The theorem does not establish that the infimum is attained, that Gaussian states are allowed, or that all states above the floor are allowed. It needs neither a Gaussian ansatz nor closure under adding Gaussian noise.

One may define $\hbar_{\mathrm{prep}} = 2\kappa_* > 0$ as a candidate reduced action scale. Identifying it with the phase constant $\hbar$ needs a separate physical argument. In quantum mechanics the familiar Robertson–Schrödinger form is $\det \mathrm{Cov}(q,p) \geq \hbar^2/4$; the equality of forms alone proves no identification. The full Planck constant would be $h = 2\pi\hbar$, not $\hbar$.

Which assumptions do the work?

Without weak closure, excluding exact points leaves preparations with arbitrarily small covariance. For example, the class of all absolutely continuous phase-space probabilities is canonically invariant, has no Dirac members, and contains arbitrarily narrow Gaussians. Its weak closure includes points. Thus the conclusion is not obtained merely by replacing points with distributions.

Without canonical squeezing, a class supported on a fixed line with a fixed nonzero extent may be weakly closed and contain no point while having zero covariance determinant. Canonical covariance rules out that degenerate evasion. Without the cusp-compatibility demand, all probabilities are weakly closed and canonically invariant and contain points; free and harmonic dynamics transport them reversibly.

Weak closure is not a consequence of finite experimental resolution. It is an additional completion rule admitting concentration limits whenever they are weak probability limits. Requiring it while retaining all arbitrarily concentrated classical preparations is exactly what would reintroduce the forbidden point. Whether nature supplies this closure, the full canonical preparation class, and the exact cusp force is open. Restricting all forces to smooth Lipschitz laws removes this particular selector.

5. A continuity selector without reversibility

The Gaussian branch-limit corollary yields a different conditional action theorem. It uses actual Newton transport of nonpoint preparations, rather than compatibility only for point preparations if any exist.

Theorem 3. Use the W_2 metric in the calibrated coordinates, and let $\mathcal{S} \subset \mathcal{P}_2(\mathbb{R}^2)$ satisfy:

1. $\mathcal{S}$ contains a state with positive-definite covariance and is invariant under affine canonical transformations.
2. $\mathcal{S}$ is closed under W_2 convergence. Independent finite products of members, followed by copy-index orthogonal canonical mixing and retention of one output pair, produce members of $\mathcal{S}$, as specified in §1 of the central-limit note.
3. At one fixed $t \in (0, (R/B)^{1/\beta})$, there is a map $\mathcal{E}_t : \mathcal{S} \to \mathcal{P}(\mathbb{R}^2)$ which is continuous from W_2 convergence of inputs to weak convergence of outputs. For every nondegenerate Gaussian member of $\mathcal{S}$, its output is the actual almost-everywhere Newton transport under the bounded cusp equation.

Then the infimum of κ on $\mathcal{S}$ is strictly positive. The map $\mathcal{E}_t$ need not be invertible, form a semigroup, or map back into $\mathcal{S}$ for this conclusion.

Proof. Work in calibrated coordinates. Suppose the infimum is zero and choose states with covariance matrices C_n and $\det C_n \to 0$. Translate their means to zero. Choose $\eta_n > 0$ such that $D_n = C_n + \eta_n I$ has $\det D_n \to 0$, and put $A_n = (\det D_n)^{1/4} D_n^{-1/2}$. In one pair $\det A_n = 1$, and

$$A_n C_n A_n^{\mathsf{T}} \le \sqrt{\det D_n}\, I. \tag{H12a}$$

The transformed admissible states have second moment tending to zero, so they converge to δ_0 in W_2; closure admits δ_0.

By the central-limit result, a positive-definite covariance member yields an admissible Gaussian with the same covariance. Translation and a determinant-one transformation give an admissible $G = \mathcal{N}(0, \zeta I)$ for some $\zeta > 0$. For each n, take one independent copy of G and $n-1$ copies of δ_0, then use an orthogonal copy-index mixer whose first row is $(1, \dots, 1)/\sqrt{n}$. The retained pair has law $\mathcal{N}(0, (\zeta/n) I)$. Translations therefore admit both Gaussian inputs (H5) with $\epsilon_n = \sqrt{\zeta/n}$ and $u = \pm 1$.

Both input sequences converge to δ_0 in W_2. The continuity assumption forces their output sequences to have the same weak limit $\mathcal{E}_t\delta_0$. Actual Newton transport and Theorem 2 give two different branch mixtures instead. This is a contradiction. □

The assumptions are consistent in a classical model. The entropy-floor class $\mathcal{E}_\zeta$ of §6 of the central-limit note has a **supplied** $\zeta > 0$, is W_2-closed, has the required canonical copy-index mixers, and is transported by the actual cusp flow. Its members are absolutely continuous. The classical flow is continuous at every initial state off the energy-zero set: bounded-force compactness and uniqueness there give continuous dependence. The exceptional set has Lebesgue measure zero, so the extended continuous-mapping theorem gives weak continuity of the transport on $\mathcal{E}_\zeta$ under W_2-convergent inputs. This provides a nonvacuous classical witness for the full continuity premise; its positive action is inherited from its imposed entropy bound.

The theorem therefore identifies a possible independently motivated stability requirement and makes its consequence explicit. It does not establish that all physical preparations obey the copy, closure and continuity rules, or that the selected floor is universal or quantum.

6. Classical witnesses and their limits

Fix a finite density ceiling $M > 0$, of reciprocal-action units, and let

$$\mathcal{S}_M = \{\mu : \ d\mu = \rho\, dq\, dp, \quad 0 \le \rho \le M, \quad \int \rho\, dq\, dp = 1\}. \tag{12}$$

This class is weakly closed: if its probabilities converge weakly, their limiting measure satisfies $\int \phi\, d\mu \le M \int \phi\, dq\, dp$ for every nonnegative compactly supported continuous ϕ, which gives the same density bound. It is invariant under affine symplectic maps because they preserve canonical area.

For (1), the energy-zero set is the union of the two graphs $p = \pm\sqrt{-2mV(q)}$ and has canonical area zero. Outside it the Newton trajectory is unique and complete. Negative-energy paths cannot reach zero; positive-energy paths cross it with nonzero velocity and are uniquely continued by $\dot q = \pm\sqrt{2(E - V(q))/m}$ near the crossing. Away from zero the field is smooth. This full-area flow preserves canonical area: where $p \neq 0$, energy–time coordinates give $dq\, dp = dE\, dt$ up to orientation, and the flow translates t; turning points away from zero have the usual smooth Hamiltonian charts. The resulting almost-everywhere group pushes $\mathcal{S}_M$ bijectively to itself. Its state curves are weakly continuous by bounded convergence along the continuous trajectories. No member charges the exceptional zero-energy set, and no Dirac preparation is admitted. Property 3 is therefore satisfied, including its stated point-compatibility condition.

The floor can be computed sharply:

$$\inf_{\mu\in\mathcal{S}_M \text{ with finite covariance}} \sqrt{\det \mathrm{Cov}_\mu(q,p)} = \frac{1}{4\pi M}. \tag{13}$$

To check it, center and canonically isotropize a positive covariance as in the reversible preparation proof, now without regularization. In the calibrated plane the density cap is MS_0. Among probabilities with this cap the minimum of $\mathbb{E}(Q^2+P^2)$ is attained by filling the disk about zero at maximum density: moving mass from a larger radius into any unfilled smaller radius reduces the expectation. The disk has radius $a_0^2 = 1/(\pi M S_0)$ and mean square radius $a_0^2/2$. Isotropy gives $2\kappa/S_0 \ge 1/(2\pi M S_0)$, proving the bound. A uniform canonical ellipse, the image of this disk, attains equality. A singular covariance would concentrate a density on a line and is impossible in (12).

Thus the conditional theorem is consistent and genuinely weaker than quantum kinematics. Its classical witness already carries a supplied scale $1/M$; the theorem says that some positive scale must appear under its hypotheses, not how nature chooses it. It is an unread-state model. Unrestricted classical conditioning can violate its ceiling, as the necessity note's two-pointer

construction proves. Claiming an operational completion would also require a physical law of instruments and records preserving its allowed states.

Composition test, 2026-10-27. The witness $\mathcal{S}_M$ is not closed under the independent-copy canonical mixer and marginalization used in the central-limit calculation. Indeed, for $f = M1_E$ with $E = -E$ and area $1/M$, the law of $(Z_1 + Z_2)/\sqrt{2}$ from independent copies has density $f_+(0) = 2\int f(z)f(-z)\,dz = 2M$, and is continuous near zero. The covariance and its action are unchanged for identical copies, but the pointwise density ceiling is violated. Theorem R and this classical single-system witness remain valid at their stated scope; they do not establish the added multi-copy closure.

7. Fine records and the action floor

Theorem 4. Let $\mathcal{S}$ be a nonempty class of Borel probabilities on the classical phase plane. Suppose that, for every $\nu \in \mathcal{S}$ and every bounded open ball D in the calibrated $(q/L_0, p/P_0)$ plane with $\nu(D) > 0$, the ordinary conditional law $\nu(\cdot \mid D)$ is again in $\mathcal{S}$. Then

$$\inf_{\nu\in\mathcal{S},\ \operatorname{Cov}\nu<\infty} \sqrt{\det \operatorname{Cov}_\nu(q,p)} = 0. \tag{H13}$$

If $\mathcal{S}$ is also weakly closed, it contains a Dirac measure.

Proof. Every probability has a point z_* in its support. The balls D_r of radius $r > 0$ centered there have positive probability. For the calibrated conditional covariance C_r, bounded support and centering at the conditional mean give $\operatorname{tr} C_r \le r^2$. Therefore

$$\kappa(\nu(\cdot \mid D_r)) = S_0\sqrt{\det C_r} \le \tfrac{1}{2}S_0 \operatorname{tr} C_r \le \tfrac{1}{2}S_0 r^2. \tag{H14}$$

These conditional laws have finite covariance and converge weakly to δ_{z_*} as $r \downarrow 0$. This proves both statements. □

The premise is an ideal, nondisturbing, arbitrarily fine *joint* record of classical position and momentum, with each positive-probability outcome admitted as a new preparation. It is not a theorem that every physical instrument has this property. Theorem I of the necessity note constructs a more specific two-pointer Hamiltonian protocol even with a prior density ceiling, but has its own explicit access and storage assumptions. In particular a prior cap alone does not supply the postulate needed to preserve a floor after records.

Combining (H13) with Theorem R shows that its nonempty, finite-covariance preparation class cannot also satisfy the fine-record postulate. Equivalently, with weak closure and affine translations, the records would admit the two point preparations that merge, contradicting its reversible phase-state evolution. This is an incompatibility among assumptions, not a proof that any one of them is physically mandatory.

8. Quantum dynamics and instruments at supplied action

For every $\hbar > 0$, the same bounded real potential has

$$H_\hbar = -\frac{\hbar^2}{2m}\frac{d^2}{dq^2} + V(q), \qquad D(H_\hbar) = H^2(\mathbb{R}). \tag{14}$$

The free operator is self-adjoint on H^2. Multiplication by V is bounded and symmetric, with relative bound zero. The Kato–Rellich theorem therefore makes (14) self-adjoint on this same domain, with $H_\hbar \ge -\|V\|_\infty$. Its spectral calculus supplies the unique strongly continuous unitary group $U_\hbar(t) = \exp(-itH_\hbar/\hbar)$ for every real time. These are [@Teschl2009, Theorem 6.4, printed p.135, and Theorem 5.1, pp.123–124; passages read in the held first edition], with the free domain in Theorem 7.8, pp.167–168. See the source companion.

This comparison needs neither an unbounded collision force nor a choice of singular boundary condition. It still supplies $\hbar > 0$; the classical density witness prevents inferring that this

is the only completion. The quantum state space also is not a class of positive joint canonical probabilities with unrestricted classical conditioning. No such identification was used in Theorem R. No relativistic premise enters, so no c is needed.

What the covariance resemblance does not imply

For every supplied physical $\hbar > 0$, the self-adjoint Hamiltonian above gives a global quantum unitary group. For a quantum state with finite position and momentum covariances, the Robertson–Schrödinger inequality gives

$$\sqrt{\det \mathrm{Cov}(Q, P)} \geq \hbar/2. \tag{Q15}$$

This remains true after any selective instrument whose conditional output is a quantum state with finite covariance. There is nevertheless an exact initial-state comparison: set

$$\epsilon_\hbar = \sqrt{\frac{\hbar}{2S_0}}, \qquad \sigma_q = L_0\epsilon_\hbar, \quad \sigma_p = P_0\epsilon_\hbar. \tag{Q15a}$$

The Gaussian in the Gaussian preparation family then has $\kappa = \hbar/2$ and is exactly the Wigner function of the normalized quantum wavefunction

$$\psi_{\hbar,u}(q) = (2\pi\sigma_q^2)^{-1/4} \exp\left[-\frac{(q - u\sigma_q)^2}{4\sigma_q^2}\right]. \tag{Q15b}$$

Thus covariance saturation and even equality of these *initial* positive phase-space functions do not by themselves select the subsequent quantum propagation or an interference law. The classical limit in Theorem 2 then corresponds to a formal $\hbar \to 0$ family at fixed S_0; no equality of the quantum and classical evolved states is asserted. At fixed physical $\hbar$, a small ratio $\hbar/S_0$ describes a large-characteristic-action regime only when L_0, P_0 specify actual scales of a family of preparations, not merely a change of units.

Equation (Q15) does not arise by conditioning a positive joint sharp-position–sharp-momentum law: the canonical Weyl operators obey

$$e^{-iaP/\hbar}e^{ibQ/\hbar} = e^{-iab/\hbar}e^{ibQ/\hbar}e^{-iaP/\hbar}, \tag{Q16}$$

and are noncommuting for general a, b. An approximate joint measurement can still have two classical outcomes; those outcomes are not the nondisturbing sharp joint record assumed in Theorem 4. The positive quantum constant in (Q15)–(Q16) is an input. Conversely, the classical history model has no noncommuting observables or demonstrated interference law, and its apparent probabilistic branch splitting in (H6) is not a coherent superposition.

There is an independently motivated empirical separator, with additional composite-system hypotheses. Under setting independence and conditional locality, an arbitrary classical hidden state including trajectory memory satisfies $|S_{\mathrm{CHSH}}| \leq 2$; the short proof and its precise assumptions are in the shared-readiness note. A Bell test with spacelike-separated electron-spin measurements reported $S_{\mathrm{CHSH}} = 2.42 \pm 0.20$ in 245 trials (Hensen et al. 2015, primary abstract read). Its setting-independence and locality interpretation is an additional physical premise, motivated by the relativistic causal setup; the recorded statistical uncertainty matters. The normalized inequality is dimensionless, so this observation does not determine an action scale or select quantum theory uniquely. Nonlocal or setting-dependent classical completions are not excluded by this bound.

9. Consequence for STATE

Reversible complete histories, reversible phase laws, continuous nonpoint cusp transport and selective-record closure are different premises. The explicit classical countermodels must accompany either conditional floor theorem. The canonical central limit proves Gaussian-sector closure under copying; the Gaussian meter separately defeats selective closure of the entropy witness and shows why a Gaussian-mixture repair fails nonlinear classical transport. The quantum preparation selection constructs semiclassical outgoing branches under supplied quantum dynamics. Those arguments have different hypotheses and remain separate. An independently justified physical continuity or instrument law, and its identification with a universal physical $\hbar$, are still missing.